

ON THE SPECTRAL STABILITY OF PERIODIC WAVES OF THE KLEIN-GORDON EQUATION

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(Submitted by: Tohru Ozawa)

Abstract. The object of study is the Klein-Gordon equation in $1 + 1$ dimensions, with integer power non-linearities. In particular, of interest is the spectral stability/instability (with respect to perturbations of the same period) of traveling-standing periodic solitons, which are of cnoidal ($p = 2$), dnoidal ($p = 3$) or more general type ($p = 5$). The corresponding linearized problem for this two-parameter family of solutions fits the general abstract framework of spectral stability for second order Hamiltonian systems, recently developed by the last two authors and Bronski-Johnson-Kapitula. It is worth noting that the spatial periodicity however, forces a relation between the speed and the phase, which results in some unique challenges in the computations of the quantities involved in the stability index. Our results generalize recent work on the simpler case of standing waves of Natali-Pastor, [9] and Natali-Cardoso, [10].

1. INTRODUCTION

Consider the following Klein-Gordon equation

$$u_{tt} - u_{xx} + u - |u|^{p-1}u = 0, x \in [-T, T], \quad (1.1)$$

AMS Subject Classifications: 35B35, 35L71, 35C07.

Accepted for publication: November 2014.

Hakkaev supported in part by research grant DDVU 02/91 of 2010 of the Bulgarian Ministry of Education and Science. Stanislavova supported in part by NSF-DMS, Applied Mathematics program, under grant # 1211315. Stefanov supported in part by NSF-DMS, Applied Mathematics program, under grant# 1313107.

where u is a complex-valued function. This equation (for various integer powers of p) is used to model physical problems of ferroelectric transitions, phase transition, crystal growth and dislocations among others, see [3]. In addition, for the case $p = 5$, this is a relativistic equation for spinless particles. These equations, together with the existence and stability of their soliton solutions¹ have been the object of numerous investigations in the last thirty years. Standing wave solutions, that is solutions in the form $e^{i\omega t}\varphi_\omega(x)$, for the higher dimensional analogues (posed on $\mathbf{R}^n$)

$$u_{tt} - \Delta u + u - F(|u|^2)u = 0, x \in \mathbf{R}^n,$$

have been studied in the seminal papers of Grillakis, [4, 5], Shatah, [12, 13] and Grillakis-Shatah-Strauss [6, 7]. In these papers, the authors established the orbital stability/instability for ground state solutions, for various values of the parameters. Identical results, but this time for spectral stability of these waves and in the full range of the parameters were established in [16], along with the stability theory for several other (not necessarily ground state) standing wave solutions. Given all the achievements that we have listed above, one can confidently say that the theory for standing waves (posed on $\mathbf{R}^n$) is well-developed and the main cases of interest are well-understood.

For periodic domains however, much less is known, even in one spatial dimension, which is our main interest. Firstly, an explicit formula for (some) solutions is available only for some integer values, like $p = 2, 3, 5$ (and possibly $p = 4$, although we do not discuss this case here). After all, lack of explicit formulas is a serious obstruction, when one studies the required spectral properties of even simple Hill operators, which is a necessity in such considerations. Next, one may consider different type of perturbations - perturbations with the same period $2T$, perturbations with period $2nT$, bounded perturbations etc. The stability of these waves of course depends on the type of these perturbations.

For the concrete case of the KG equation (1.1), Natali and Pastor have considered standing waves for the cubic KG (i.e., $p = 3$) both in the focusing and defocusing cases. In [9], they have considered the case $p = 3$ and the corresponding cnoidal and dnoidal solutions in the focussing case, and the snoidal solutions in the defocussing case. For certain values of the parameters, they have established an orbital stability result for the dnoidal solutions and orbital instability of the cnoidal and snoidal solutions. For the quintic case, $p = 5$, Natali and Cardoso, [10] have constructed periodic standing wave solutions and they have shown orbital instability for such

¹which provide important information for the underlying physical models

waves in certain regime of the parameters (thus matching the instability results from the aforementioned works on the whole line case). Both of these works have utilized the Grillakis-Shatah-Strauss machinery and in addition, *the perturbations are taken to be even functions and with the same period.*

In this paper, we are interested in the stability of a two-parameter family of solutions, namely the traveling-standing wave solutions of (1.1). More precisely, we seek solutions in the form

$$e^{i\omega t} e^{iq(x-ct)} \varphi_{\omega,c}(x-ct), \quad (1.2)$$

for some real parameters ω, c, q and a real-valued function φ , which depends on these parameters.

On the real line, it is easy to show that explicit *sech* type solutions exists if $\omega^2 + c^2 < 1$ and for any real $p > 1$. The stability of these waves has been investigated, in the whole line context, in the recent paper by Stanislavova, [14]. It was shown that they are spectrally stable if and only if $1 < p \leq 1 + \frac{4\omega^2}{1-c^2}$. Note in particular that for $p \geq 5$, one has instability for all values of ω, c .

The purpose of this paper is to extend these results to the periodic context, at least for some integer values of p . The paper is organized as follows. In Section 2, we give a detailed derivation of the formulas for the solutions that we work with, in all cases $p = 2, 3, 5$. In Section 3, we set up the linearized problem and give the general abstract result that we use for second order in time Hamiltonian systems. Subsequently, we verify the spectral assumptions, required by the abstract results. Section 4 is devoted to the specific computation of the index that gives the condition for stability (or the instability) of the corresponding waves, (4.17).

2. CONSTRUCTION OF PERIODIC STANDING-TRAVELLING WAVES

Plugging in the ansatz (1.2) in (1.1) yields the ODE

$$(c^2 - 1)\varphi'' + 2i(c(\omega + qc) - q)\varphi' + (q^2 - (\omega + qc)^2 + 1)\varphi - \varphi^p = 0, \quad x \in [-T, T].$$

In order to have real solution φ , we must then require $c(\omega + qc) - q = 0$ or equivalently, $q = \frac{\omega c}{1-c^2}$. Note that the periodicity in the x variable now requires $\varphi(-T) = \varphi(T)$ and in addition²,

$$\frac{\omega c}{1-c^2} = q = \frac{\pi n}{T}, n \in \mathbf{Z}. \quad (2.1)$$

²For the purposes of this paper, we will assume that $n \neq 0$. Indeed, $n = 0$, will reduce us to either $\omega = 0$ or $c = 0$, which are considerably simpler problem to consider. In addition, some of these cases were considered in previous publications like [9], [10]

The defining equation for φ becomes

$$-(1 - c^2)\varphi'' + \frac{1 - \omega^2 - c^2}{1 - c^2}\varphi - \varphi^p = 0, \quad x \in [-T, T]. \quad (2.2)$$

In order to have existence, we will be assuming that $1 > \omega^2 + c^2$. With this assumption in mind, we introduce the positive parameters

$$\mu := \sqrt{1 - c^2} > 0, \quad \nu := \frac{1 - \omega^2 - c^2}{1 - c^2} > 0,$$

so that (2.2) now reads

$$-\mu^2\varphi'' + \nu\varphi - \varphi^p = 0, \quad x \in [-T, T]. \quad (2.3)$$

Integrating once the above equation, we get

$$-\mu^2\varphi'^2 + \nu\varphi^2 - \frac{2}{p+1}\varphi^{p+1} = a, \quad (2.4)$$

where a is a constant of integration and φ is periodic provided that the level set $H(X, Y) = a$ of the Hamiltonian system $dH = 0$, with

$$H(X, Y) = Y^2 - \frac{\nu}{\mu^2}X^2 + \frac{2}{\mu^2(p+1)}X^{p+1},$$

contains a periodic trajectory (an oval).

2.1. The case $p = 2$. In this case, the level set $H(X, Y) = a$ contains two periodic trajectories if $\mu > 0$, $a \in (-\frac{1}{3}\mu^3, 0)$ and a unique periodic trajectory if $\mu \in \mathbb{R}$, $a > 0$. Under these conditions, equation (2.4) becomes $H(\varphi, \varphi') = a$ and its solution φ is periodic of period $T = T(c, a)$.

Let $a < 0$ ($c > 0$). Then either $\varphi < 0$ (the left case) or $\varphi > 0$ (the right case). We will consider the right case $\varphi > 0$. Denote by $\varphi_0 > \varphi_1 > 0$ the positive solutions of $-\frac{2}{3\mu^2}\rho^3 + \frac{\nu}{\mu^2}\rho^2 - a = 0$. Then $\varphi_1 \leq \varphi \leq \varphi_0$ and one can rewrite (2.4) as

$$\varphi'^2 = \frac{2}{3\mu^2}(\varphi - \varphi_1)(\varphi_0 - \varphi)(\varphi + \varphi_0 + \varphi_1 - \frac{3\nu}{2}). \quad (2.5)$$

Therefore, $2\varphi_0 + \varphi_1 > \varphi_0 + 2\varphi_1 > \frac{3\nu}{2}$. Introducing a new variable $s \in (0, 1)$ via $\varphi = \varphi_1 + (\varphi_0 - \varphi_1)s^2$, we transform (2.5) into

$$s'^2 = \alpha^2(1 - s^2)(k'^2 + k^2s^2),$$

where α, k, k' are positive constants ($k^2 + k'^2 = 1$) given by

$$\alpha^2 = \frac{4\varphi_0 + 2\varphi_1 - 3\nu}{12\mu^2}, \quad k^2 = \frac{2\varphi_0 - 2\varphi_1}{4\varphi_0 + 2\varphi_1 - 3\nu}, \quad k'^2 = \frac{2\varphi_0 + 4\varphi_1 - 3\nu}{4\varphi_0 + 2\varphi_1 - 3\nu}. \quad (2.6)$$

Therefore, up to translation we have (see Chapter 22, [17]) that

$$\varphi = \varphi_1 + (\varphi_0 - \varphi_1)cn^2(\alpha x; k). \quad (2.7)$$

Since the period of the elliptic function $cn^2(x; \kappa)$ is $2K(\kappa)$, then for the fundamental period of standing wave solution φ we get

$$T = \frac{K(k)}{\alpha} = \frac{2\mu\sqrt{1-k^2+k^4}K(k)}{\sqrt{\nu}}, \quad k \in (0, 1), \quad T \in I = \left(\frac{2\pi}{\sqrt{\nu}}, \infty\right). \quad (2.8)$$

To express c through α and k , one should take into account the fact that in the cubic equation we used to determine φ_0 and φ_1 , the coefficient at ρ is zero. Therefore,

$$\varphi_0\varphi_1 + (\varphi_0 + \varphi_1)\left(\frac{3\nu}{2} - \varphi_0 - \varphi_1\right) = 0,$$

and

$$\begin{aligned} \varphi_0 - \varphi_1 &= 6\mu^2\alpha^2\kappa^2, \\ \varphi_0 &= 2\mu^2\alpha^2(1 + \kappa^2) + \frac{1}{2}\nu, \quad \varphi_1 = 2\mu^2\alpha^2(1 - 2\kappa^2) + \frac{1}{2}\nu \\ \nu^2 &= 16\mu^4\alpha^4(1 - \kappa^2 + \kappa^4). \end{aligned} \quad (2.9)$$

2.2. The case $p = 3$. In this case, we have

$$H(X, Y) = Y^2 - \frac{\nu}{\mu^2}X^2 + \frac{1}{2\mu^2}X^4.$$

The level set $H(x, y) = a$ contains two periodic trajectories if $\nu > 0$, $a \in (-\frac{1}{2}\nu^2, 0)$ and a unique periodic trajectory if $\nu \in \mathbb{R}$, $a > 0$. Under these conditions, the solution of (2.4) is determined by $H(\varphi, \varphi') = a$ and φ is periodic of period $T = T(\mu, \nu, a)$.

Let $\nu > 0$ and $\varphi_0 > \varphi_1$ are the positive roots of $-\frac{1}{2\mu^2}\rho^4 + \frac{\nu}{\mu^2}\rho^2 - a = 0$. Then, up to translation (see Chapter 22, [17]), we obtain the explicit formulas

$$\varphi(x) = \varphi_0 \operatorname{dn}(\alpha x; \kappa), \quad (2.10)$$

and

$$\kappa^2 = \frac{\varphi_0^2 - \varphi_1^2}{\varphi_0^2} = \frac{2\varphi_0^2 - 2\nu}{\varphi_0^2}, \quad \alpha = \frac{\varphi_0}{\sqrt{2\mu}}, \quad T = \frac{K(\kappa)}{\alpha}. \quad (2.11)$$

2.3. The case $p = 5$. In this case we consider the positive solution of the level set $H(x, y) = a$. Let $\varphi = \sqrt{\phi}$. Then for ϕ , we get

$$\phi'^2 = \frac{4}{3\mu^2}(-\phi^4 + 3\nu\phi^2 + 3a\phi). \quad (2.12)$$

Let $\phi_1 < \phi_2 < \phi_3$ are roots of the polynomial $F(s) = -s^3 + 3\nu s + 3a$. We have

$$\phi'^2 = \frac{4}{3}\phi(\phi - \phi_1)(\phi - \phi_2)(\phi_3 - \phi), \quad (2.13)$$

and

$$\phi_1 + \phi_2 + \phi_3 = 0, \quad \phi_1\phi_2 + \phi_1\phi_3 + \phi_2\phi_3 = -3\nu, \quad \phi_1\phi_2\phi_3 = 3a. \quad (2.14)$$

From the relations (2.14) we have $\nu > 0$ and $\phi_1 < 0 < \phi_2 < \phi_3$, $\phi_2 < \phi < \phi_3$. Integrating the equation (2.13), then up to translation we get

$$\int_{\phi}^{\phi_3} \frac{ds}{\sqrt{s(s-\phi_1)(s-\phi_2)(\phi_3-s)}} = \frac{2}{\sqrt{3\mu g}} \xi,$$

and the solution ϕ [see [2]] is given by

$$\phi(\xi) = \frac{\phi_3(\phi_2 - \phi_1) + \phi_1(\phi_3 - \phi_2)sn^2(\frac{2}{\sqrt{3\mu g}}, \kappa)}{(\phi_2 - \phi_1) + (\phi_3 - \phi_2)sn^2(\frac{2}{\sqrt{3\mu g}}, \kappa)}, \quad (2.15)$$

where $g = \frac{2}{\sqrt{\phi_3(\phi_2 - \phi_1)}}$ and $\kappa^2 = \frac{-\phi_1(\phi_3 - \phi_2)}{\phi_3(\phi_2 - \phi_1)}$. The fundamental period of the function (2.15) is given by

$$T = \sqrt{3\mu g}K(\kappa) = \frac{2\sqrt{3}\mu K(\kappa)}{\sqrt[4]{12\nu\phi_3^2 - 3\phi_3^4}}. \quad (2.16)$$

Further, we have the following relations

$$\begin{aligned} \nu &= \frac{4\mu^2 K^2(\kappa)}{T^2} \sqrt{\kappa^4 - \kappa^2 + 1}, \\ \phi_3^2 &= \frac{4\mu^2 K^2(\kappa)}{T^2} \left[2\sqrt{\kappa^4 - \kappa^2 + 1} + (2\kappa^2 - 1) \right], \quad g = \frac{T}{\sqrt{3\mu K(\kappa)}}. \end{aligned} \quad (2.17)$$

From the above relations, for the nominator in explicit form of the function ϕ , we have

$$\begin{aligned} &\phi_3(\phi_2 - \phi_1) + \phi_1(\phi_3 - \phi_2)sn^2\left(\frac{2}{\sqrt{3\mu g}}\xi, \kappa\right) \\ &= 12\mu^2 \frac{K^2(\kappa)}{T^2} dn^2\left(\frac{2}{\sqrt{3\mu g}}\xi, \kappa\right) = M(\xi, \kappa), \end{aligned}$$

and for the denominator, we have

$$\begin{aligned} &(\phi_2 - \phi_1) + (\phi_3 - \phi_2)sn^2\left(\frac{2}{\sqrt{3\mu g}}\xi, \kappa\right) \\ &= 2\sqrt{3}\frac{\mu K(\kappa)}{T} \sqrt{2\sqrt{\kappa^4 - \kappa^2 + 1} - (2\kappa^2 - 1)} \\ &+ 3\frac{\mu K(\kappa)}{T} \sqrt{2\sqrt{\kappa^4 - \kappa^2 + 1} + (2\kappa^2 - 1)} sn^2\left(\frac{2}{\sqrt{3\mu g}}\xi, \kappa\right) - \\ &- \sqrt{3}\frac{\mu K(\kappa)}{T} \sqrt{2\sqrt{\kappa^4 - \kappa^2 + 1} - (2\kappa^2 - 1)} sn^2\left(\frac{2}{\sqrt{3\mu g}}\xi, \kappa\right) = N(\xi, \kappa). \end{aligned}$$

Finally, for the solution φ , we have $\varphi(x, \kappa) = \sqrt{\frac{M(x, \kappa)}{N(x, \kappa)}}$.

3. LINEARIZED EQUATION: THE SPECTRAL STABILITY PROBLEM

In this section, we set up the linearized problem for (1.1). We show that the problem reduces to a specific form of a quadratic operator pencil, for which a general theory has been developed in the recent papers [15], [16], [1].

3.1. Setting up the linearized problem. To this end, we take the perturbation in the form $u = e^{i\omega t} e^{iq(x-ct)} (\varphi(x-ct) + v(t, x-ct))$. Plugging this ansatz in (1.1) and ignoring all quadratic and higher order terms yields the following *linear* equation for v

$$v_{tt} - 2cv_{tx} - 2i(\omega + cq)v_t + (-\mu^2 \partial_x^2 + \nu - \varphi^{p-1})v - (p-1)\varphi^{p-1}\Re v = 0.$$

Splitting the real and imaginary parts $v = w + iz$ allows us to rewrite the linearized problem as the following system

$$\begin{cases} w_{tt} - 2cw_{tx} + 2(\omega + cq)z_t + L_+ w = 0 \\ z_{tt} - 2cz_{tx} - 2(\omega + cq)w_t + L_- z = 0, \end{cases} \quad (3.1)$$

where

$$L_+ = -\mu^2 \partial_x^2 + \nu - p\varphi^{p-1}, \quad L_- = -\mu^2 \partial_x^2 + \nu - \varphi^{p-1}.$$

With these assignments, the linearized problem can be written as

$$\begin{pmatrix} w \\ z \end{pmatrix}_{tt} + 2 \begin{pmatrix} -c\partial_x & (\omega + qc) \\ -(\omega + qc) & -c\partial_x \end{pmatrix} \begin{pmatrix} w \\ z \end{pmatrix}_t + \begin{pmatrix} L_+ & 0 \\ 0 & L_- \end{pmatrix} \begin{pmatrix} w \\ z \end{pmatrix} = 0. \quad (3.2)$$

Thus, we may rewrite it further as

$$\mathbf{v}_{tt} + \mathcal{J}\mathbf{v}_t + \mathcal{H}\mathbf{v} = 0, \quad (3.3)$$

where

$$\mathbf{v} = \begin{pmatrix} w \\ z \end{pmatrix}, \quad \mathcal{H} = \begin{pmatrix} L_+ & 0 \\ 0 & L_- \end{pmatrix}, \quad \mathcal{J} = 2 \begin{pmatrix} -c\partial_x & (\omega + qc) \\ -(\omega + qc) & -c\partial_x \end{pmatrix}.$$

Note that $\mathcal{H}^* = \mathcal{H}$, while $\mathcal{J}^* = -\mathcal{J}$. If we consider the eigenvalue problem associated with (3.3), that is $\mathbf{v} = e^{\lambda t}\psi$, we arrive at

$$\lambda^2 \psi + \mathcal{J}\lambda \psi + \mathcal{H}\psi = 0. \quad (3.4)$$

The form (3.4) of the linearized problem fits the theory developed in [1]. Before we present the results of [1], let us give the following definition.

Definition 1. We say that the quadratic pencil given by the couple (J, H) is spectrally unstable, if there exists a T periodic function $\psi \in D(\mathcal{H})$ and $\lambda : \mathbb{R} \lambda > 0$, so that

$$\lambda^2 \psi + \lambda J \psi + H \psi = 0.$$

Otherwise, we say that the quadratic pencil (J, H) is spectrally stable.

3.2. Spectral stability of quadratic pencils. We now present the theory developed in [1] for the stability of quadratic pencils, which we will be able to apply to our problem (3.4). We only present a corollary of the results therein, which fits our purposes. We start with some notations. For a self-adjoint operator H , define the number of the negative eigenvalues

$$n(H) = \#\{\lambda \in (-\infty, 0) \cap \sigma(H)\}.$$

Next, for a subspace M of a Hilbert space L , denote $P_M : L \rightarrow M$ to be the orthogonal projection onto the subspace M and $P_M^\perp := Id - P_M$. For an operator T , denote $T_M := P_M T P_M$. In particular, if M is finite dimensional, with orthogonal basis $\{\chi_j\}_{j=1}^n$, T_M is given in a matrix form by $\{\langle T \chi_i, \chi_j \rangle\}_{i,j=1}^n$.

Theorem 1. ([1]) Let $H = H^*$, so that $n(H) < \infty$ and $\dim(Ker(H)) < \infty$. Denote $P_H := P_{Ker(H)}$. Assume that

$$(1) \quad (P_H^\perp H P_H^\perp)^{-1}, (P_H^\perp H P_H^\perp)^{-1} P_H^\perp J P_H^\perp, \quad (3.5)$$

are all compact operators on $L^2[-T, T]$.

- (2) There exists a positive operator S , so that $X_S = \{u : \langle Su, Su \rangle < \infty\}$ is dense in L^2 .
- (3) $S^{-1} J S^{-1}, S^{-1}$ are compact, whereas $S(H + \lambda)^{-1} S$ is compact for $\lambda \gg 1$.
- (4) $J : Ker(H) \rightarrow Ker(H)^\perp$.
- (5) $(Id - JH^{-1}J)|_{Ker(H)}$ is invertible³.

Then,

$$k_r + k_c + k_- = n(H) - n((Id - JH^{-1}J)|_{Ker(H)}), \quad (3.6)$$

where k_r is the number of positive solutions λ of (3.4), k_c is the number of solutions λ of (3.4) with positive real part, whereas k_- is the total negative Krein index for the quadratic pencil⁴.

³Note that this is well-defined since $J : Ker(H) \rightarrow Ker(H)^\perp$ and hence $JH^{-1}J$ is well-defined.

⁴This is a technical term that appears in the count, which is the number of purely imaginary eigenvalues satisfying certain technical condition. The important thing for us that if $i\mu$ is such an e-value, so is $-i\mu$ and hence k_- is an even number

Remark. The non-negative integers k_c, k_- are even. As a consequence, if $n(H) = 1$, it follows from (3.6) that $k_c = k_- = 0$ and

$$k_r = 1 - n((Id - JH^{-1}J)|_{Ker(H)}). \quad (3.7)$$

3.3. The pencil $(\mathcal{J}, \mathcal{H})$ satisfies Theorem 1. We now verify that the pencil given by $(\mathcal{J}, \mathcal{H})$ (or equivalently the eigenvalue problem (3.4)) satisfy the requirements of Theorem 1. The compactness of the operators in (3.5) follows from the compactness of the embeddings $H^2[-T, T] \hookrightarrow H^1[-T, T] \hookrightarrow L^2[-T, T]$ as follows

$$(P_H^\perp H P_H^\perp)^{-1} : L^2 \rightarrow H^2 \hookrightarrow L^2, \quad (P_H^\perp H P_H^\perp)^{-1} P_H^\perp J P_H^\perp : L^2 \rightarrow H^1 \hookrightarrow L^2.$$

We take the operator $S := (1 + |\partial_x|)^{3/4}$. Clearly $X_S = H^{3/4}[-T, T]$ is dense in L^2 . In addition, $S^{-1}JS^{-1}$ is smoothing of order $1/2$, S^{-1} smoothing of order $3/4$, whereas $S(H + \lambda)^{-1}S, \lambda \gg 1$ is smoothing of order $1/2$. Thus, all the operators in question are compact in their action on L^2 .

Next, we list the spectral properties of the operator $\mathcal{H}$. Since it is a diagonal operator, with entries $L_\pm$, we have that $\sigma(\mathcal{H}) = \sigma(L_+) \cup \sigma(L_-)$. The operators $L_\pm$ are clearly of Hill type, so that the standard Sturm-Liouville theory is applicable to them. In particular, since for all cases $p = 2, 3, 5$ we have that $L_-(\varphi) = 0$ and $\varphi > 0$, it follows that $\sigma(L_-) \subset [0, \infty)$. In addition, by differentiating in x the defining equation (2.3), we obtain that $L_+[\varphi'] = 0$, whence 0 is an eigenvalue of multiplicity at least two for $\mathcal{H}$.

In fact, the spectrum of both $L_\pm$ is well-known in the cases $p = 2, 3$ and $\dim(Ker(\mathcal{H})) = 2$, with $Ker(\mathcal{H})$ spanned by the eigenfunctions that we have already found, namely φ, φ' . Denote the normalized elements of the kernel by ψ_0, ψ_1 . That is,

$$\psi_0 = \begin{pmatrix} 0 \\ \varphi \|\varphi\|^{-1} \end{pmatrix}, \quad \psi_1 = \begin{pmatrix} \varphi' \|\varphi'\|^{-1} \\ 0 \end{pmatrix}.$$

Next, we verify by inspection

$$\langle \mathcal{J}\psi_0, \psi_1 \rangle = \frac{2(\omega + qc)}{\|\varphi\| \|\varphi'\|} \langle \varphi, \varphi' \rangle = 0.$$

The anti-selfadjointness of $\mathcal{J}$ yields $\langle \mathcal{J}\psi_i, \psi_i \rangle = 0, i = 1, 2$. With this, we have verified that $\mathcal{J} : Ker(H) \rightarrow Ker(H)^\perp$.

We now review the known spectral results for L_+ in the cases $p = 2, 3$. These will imply that L_+ has one simple negative eigenvalue (when $p = 2, 3$), whence $n(\mathcal{H}) = n(L_+) = 1$, hence formula (3.7) will be applicable.

In the case $p = 2$, using that $sn^2(y) + cn^2(y) = 1$ and (2.9), we obtain

$$\begin{aligned} L_+ &= -\mu^2 \partial_x^2 + \nu - 2\varphi_1 - 2(\varphi_0 - \varphi_1)cn^2(\alpha x, \kappa) \\ &= \alpha^2 \mu^2 [-\partial_y^2 + 12\kappa^2 sn^2(y, \kappa) - 4(1 + \kappa^2)], \end{aligned}$$

where $y = \alpha x$. The spectral properties of the operator $\Lambda_1 = -\frac{d^2}{dy^2} - 4(1 + \kappa^2) + 12\kappa^2 sn^2(y; \kappa)$ in $[0, 2K(k)]$ are well known [8]. The first three eigenvalues are simple and zero is the second one. Since the eigenvalues of L_1 and Λ_1 are related by $\lambda_n = \alpha^2 \mu_n$, it follows that the first three eigenvalues of the operator L_+ , equipped with periodic boundary condition on $[0, 2K(k)]$ are simple and the first eigenvalue is negative and zero is the second eigenvalue.

Similarly, when $p = 3$ using that $\kappa^2 sn^2(y) + dn^2(y) = 1$ and by (2.11), we obtain

$$\begin{aligned} L_+ &= -\mu^2 \partial_x^2 + \mu^2 \alpha^2 (2 - \kappa^2) - 6\mu^2 \alpha^2 (1 - \kappa^2 sn^2(\alpha x, \kappa)) \\ &= \alpha^2 \mu^2 [-\partial_y^2 + 6\kappa^2 sn^2(y, \kappa) - (4 + \kappa^2)]. \end{aligned}$$

The spectral properties of the operator $\Lambda_2 = -\frac{d^2}{dy^2} - (4 + \kappa^2) + 6\kappa^2 sn^2(y; \kappa)$ in $[0, 2K(k)]$ are also known, [8]. The first three eigenvalues are simple and zero is the second one. Since the eigenvalues of L_+ and Λ_2 are related by $\lambda_n = \alpha^2 \mu_n$, it follows that the first three eigenvalues of the operator L_+ , equipped with periodic boundary condition on $[0, 2K(k)]$ are simple and the first eigenvalue is negative and zero is the second eigenvalue.

Thus, we have verified that for the cases $p = 2, 3$, the pencil $(\mathcal{J}, \mathcal{H})$ satisfies the assumptions of Theorem 1, modulo the conditions for the invertibility of $(Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})|_{Ker(\mathcal{H})}$. This will be postponed for the next section, where the value of the index $n((Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})|_{Ker(\mathcal{H})})$ will be explicitly calculated in terms of the parameters involved.

3.4. The case $p = 5$. When $p = 5$, we still have that $L_- \geq 0$, with one simple eigenvalue at zero. The question for the spectrum of L_+ is more complicated, see [10]. In it, the authors have developed, based in parts on an earlier work of Neves [11], a criteria to determine⁵ the multiplicity of the zero eigenvalue as well as the dimension of the negative subspace of L_+ . By Sturm-Liouville theory (see Theorems 3.4 and 3.5 in [10]) and since φ' has exactly two zeroes in $[-T, T]$, it follows that several configurations are possible: $n(L_+) = 1$ and $\dim(Ker(L_+)) = 1$ or $n(L_+) = 1$, $\dim(Ker(L_+)) = 2$ or $n(L_+) = 2$, $\dim(Ker(L_+)) = 1$.

⁵recall however that this applies to the spectrum of L_+ , when the stationary solution is a standing wave - in our notations $c = 0$

Natali and Cardoso's have (numerically) showed that for all $\omega \in (0, 1)$ (recall $c = 0$), it is the case that $\dim(\text{Ker}(L_+)) = 1$ and $n(L_+) = 1$. Our results below, in the case $p = 5$, will also be relative to the spectral hypothesis

$$\text{Ker}(L_+) = \text{span}\{\varphi'\}, \quad n(L_+) = 1, L_- \geq 0, 0 \text{ is simple e-value}, \quad (3.8)$$

which is satisfied (in the case $c = 0$) for all $\omega \in (0, 1)$. In addition to the numerical results in [10], we have prepared our own numerical runs, which clearly show that (3.8) is satisfied for all values of ω , see Figure 1 below for more details.

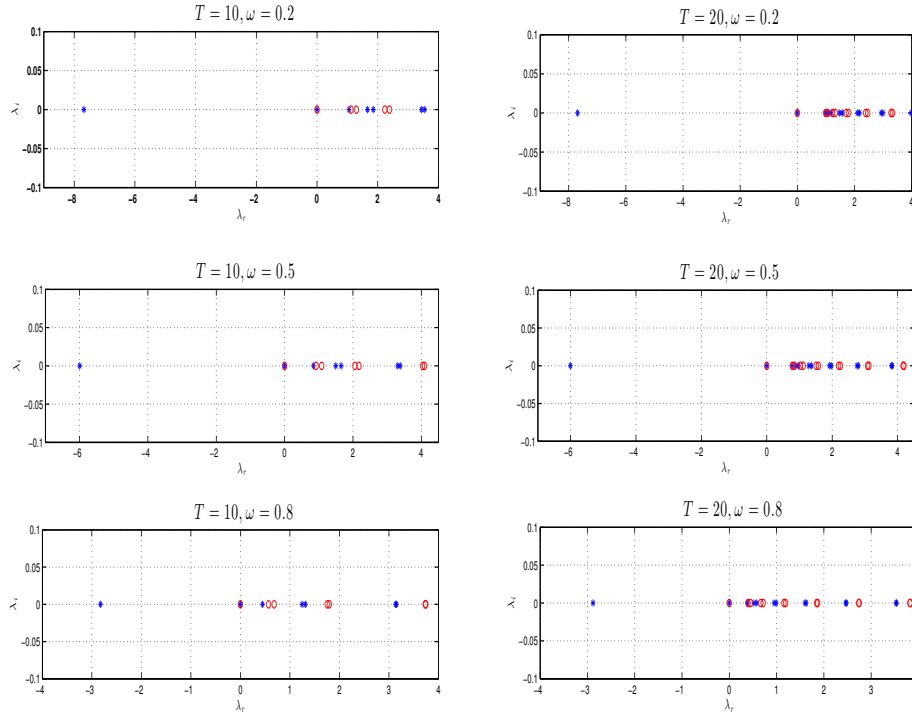


FIGURE 1. Each panel shows the spectral plane (λ_r, λ_i) of the eigenvalues $\lambda = \lambda_r + i\lambda_i$ for the linearization around the standing wave solution for the discrete case for a specific T and ω value. The blue stars represent the eigenvalues of L_+ and the red circles represent the eigenvalues of L_- . Extensive numerical runs over all values of T, ω have shown the same structure of the spectrum of H - a double eigenvalue at zero and a simple negative e-value.

4. CALCULATION OF THE INDEX

We need to consider the matrix representation of $(Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})|_{Ker(\mathcal{H})}$. It is the symmetric matrix

$$\begin{aligned} U &:= \begin{pmatrix} \langle (Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})\psi_0, \psi_0 \rangle & \langle (Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})\psi_0, \psi_1 \rangle \\ \langle (Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})\psi_1, \psi_0 \rangle & \langle (Id - \mathcal{J}\mathcal{H}^{-1}\mathcal{J})\psi_1, \psi_1 \rangle \end{pmatrix} \\ &= \begin{pmatrix} 1 + \langle \mathcal{H}^{-1}\mathcal{J}\psi_0, \mathcal{J}\psi_0 \rangle & \langle \mathcal{H}^{-1}\mathcal{J}\psi_0, \mathcal{J}\psi_1 \rangle \\ \langle \mathcal{H}^{-1}\mathcal{J}\psi_1, \mathcal{J}\psi_0 \rangle & 1 + \langle \mathcal{H}^{-1}\mathcal{J}\psi_1, \mathcal{J}\psi_1 \rangle \end{pmatrix}. \end{aligned}$$

Recall that we need to compute the number of negative eigenvalues of this matrix U . In view of the formula (3.7), U cannot have two negative eigenvalues. Hence, the stability criteria may be formulated as follows

- Stability, if $\det(U) < 0$ (one positive and one negative eigenvalue, hence $n(U) = 1$),
- Instability, if $\det(U) > 0$ (two positive eigenvalues, $n(U) = 0$).

Thus, it remains to compute $\det(U)$. To this end,

$$\begin{aligned} \mathcal{H}^{-1}\mathcal{J}\psi_0 &= 2\|\varphi\|^{-1}\mathcal{H}^{-1} \begin{pmatrix} \frac{\omega}{1-c^2}\varphi \\ -c\varphi' \end{pmatrix} = 2\|\varphi\|^{-1} \begin{pmatrix} \frac{\omega}{1-c^2}L_+^{-1}[\varphi] \\ -cL_-^{-1}[\varphi'] \end{pmatrix} \\ \mathcal{H}^{-1}\mathcal{J}\psi_1 &= 2\|\varphi'\|^{-1}\mathcal{H}^{-1} \begin{pmatrix} -c\varphi'' \\ -\frac{\omega}{1-c^2}\varphi' \end{pmatrix} = 2\|\varphi'\|^{-1} \begin{pmatrix} -cL_+^{-1}[\varphi''] \\ -\frac{\omega}{1-c^2}L_-^{-1}[\varphi'] \end{pmatrix}. \end{aligned}$$

Thus,

$$\begin{aligned} \langle \mathcal{H}^{-1}\mathcal{J}\psi_0, \mathcal{J}\psi_0 \rangle &= 4\|\varphi\|^{-2} \left(\frac{\omega^2}{(1-c^2)^2} \langle L_+^{-1}[\varphi], \varphi \rangle + c^2 \langle L_-^{-1}[\varphi'], \varphi' \rangle \right) \\ \langle \mathcal{H}^{-1}\mathcal{J}\psi_1, \mathcal{J}\psi_0 \rangle &= \langle \mathcal{H}^{-1}\mathcal{J}\psi_0, \mathcal{J}\psi_1 \rangle \\ &= 4\|\varphi\|^{-1}\|\varphi'\|^{-1} \frac{\omega c}{1-c^2} \left(\langle L_-^{-1}[\varphi'], \varphi' \rangle - \langle L_+^{-1}[\varphi], \varphi'' \rangle \right) \\ \langle \mathcal{H}^{-1}\mathcal{J}\psi_1, \mathcal{J}\psi_1 \rangle &= 4\|\varphi'\|^{-2} \left(c^2 \langle L_+^{-1}[\varphi''], \varphi'' \rangle + \frac{\omega^2}{(1-c^2)^2} \langle L_-^{-1}[\varphi'], \varphi' \rangle \right). \end{aligned}$$

From these expressions, it is clear that it remains to compute the expressions $\|\varphi\|$, $\|\varphi'\|$, $\langle L_-^{-1}[\varphi'], \varphi' \rangle$ and various quantities involving $L_+^{-1}[\varphi]$, $L_+^{-1}[\varphi'']$.

4.1. Computation of $\langle L_-^{-1}[\varphi'], \varphi' \rangle$. From (2.3), we have $L_-\varphi = 0$. The function $\phi = \varphi(x) \int_0^x \frac{1}{\varphi^2(y)} dy$ is also solution of $L_-\phi = 0$, which is independent of φ . Thus, we may construct the Green's function as follows

$$L_-^{-1}f = \frac{\varphi(x)}{\mu^2} \int_0^x \phi(y)f(y)dy - \frac{\phi(x)}{\mu^2} \int_0^x \varphi(y)f(y)dy + C_f\phi(x), \quad (4.1)$$

where the constant C_f is chosen such that $L_-^{-1}f$ is periodic with same period as $\varphi(x)$. From (4.1) for $L_-^{-1}\varphi'$, we get

$$L_-^{-1}\varphi' = \frac{\varphi(x)}{\mu^2} \int_0^x \phi(y)\varphi'(y)dy - \frac{\phi(x)}{\mu^2} \int_0^x \varphi(y)\varphi'(y)dx + C_f\phi(x), \quad (4.2)$$

and for constant C_f , we have

$$C_{\varphi'} = -\frac{\varphi(T)}{2\mu^2\phi(T)}\langle\varphi',\phi\rangle + \frac{1}{2\mu^2}(\varphi^2(T) - \varphi^2(0)). \quad (4.3)$$

For the expression $\langle L_-^{-1}[\varphi'], \varphi' \rangle$, we get

$$\begin{aligned} \langle L_-^{-1}[\varphi'], \varphi' \rangle &= -\frac{1}{\mu^2}\langle\varphi^2\varphi', \phi\rangle + \frac{1}{\mu^2}\varphi^2(T)\langle\varphi', \phi\rangle - \frac{\varphi(T)}{2\mu^2\phi(T)}\langle\varphi', \phi\rangle^2 \\ &= -\frac{1}{\mu^2}I_{11} + \frac{1}{\mu^2}\varphi^2(T)I_{12} - \frac{\varphi(T)}{2\mu^2\phi(T)}I_{12}^2, \end{aligned} \quad (4.4)$$

where

$$I_{11} = \langle\varphi^2\varphi', \phi\rangle = \frac{1}{4}\left[\varphi^4(T) \int_{-T}^T \frac{1}{\varphi^2}dx - \int_{-T}^T \varphi^2dx\right], \quad (4.5)$$

and

$$I_{12} = \langle\varphi', \phi\rangle = \frac{1}{2}\left[\varphi^2(T) \int_{-T}^T \frac{1}{\varphi^2}dx - 2T\right]. \quad (4.6)$$

4.2. Computation of $\langle L_+^{-1}[\varphi], \varphi \rangle, \langle L_+^{-1}[\varphi], \varphi'' \rangle, \langle L_+^{-1}[\varphi''], \varphi'' \rangle$. For this part, we need to analyze the defining equation (2.3). Before that, let us record the relation between μ, ν , which is forced by (2.1). Namely, after resolving (2.1) for ν , we obtain

$$\nu = \nu_n(\mu, T) = 1 + \frac{\mu^2}{\mu^2 - 1} \left(\frac{\pi n}{T}\right)^2. \quad (4.7)$$

Next, for (2.3), we write $\varphi = \varphi(T, \mu; x) = \varphi(T, \mu; Ty)$, where $y \in [-1, 1]$. Upon introducing $\tilde{\varphi}(T, \mu; y) := \varphi(T, \mu; Ty)$, we can rewrite (2.3) as

$$-\mu^2 T^{-2} \tilde{\varphi}''(T, \mu; y) + \nu_n(\mu, T) \tilde{\varphi}(T, \mu; y) - \tilde{\varphi}^p(T, \mu; y) = 0, \quad -1 \leq y \leq 1. \quad (4.8)$$

Introduce

$$\tilde{L}_+ := -\mu^2 T^{-2} \partial_{yy} + \nu_n(\mu, T) - p\tilde{\varphi}^{p-1}.$$

Note that by a simple scaling argument,

$$\text{Ker}(\tilde{L}_+) = \text{span}[\tilde{\varphi}'],$$

since $\text{Ker}[L_+] = \text{span}[\varphi']$. We now have two independent parameters to play with in (4.8), namely T, μ . If we take a derivative with respect to μ , we obtain

$$\tilde{L}_+ \left[\frac{\partial \tilde{\varphi}}{\partial \mu} \right] - 2\mu T^{-2} \tilde{\varphi}'' + \frac{\partial \nu_n(\mu, T)}{\partial \mu} \tilde{\varphi} = 0,$$

Inverting the operator $\tilde{L}_+$ yields

$$2\mu T^{-2} \tilde{L}_+^{-1}[\tilde{\varphi}'] - \frac{\partial \nu_n(\mu, T)}{\partial \mu} \tilde{L}_+^{-1}[\tilde{\varphi}] = \frac{\partial \tilde{\varphi}}{\partial \mu} + c_1(\mu, T) \varphi',$$

or

$$2\mu \tilde{L}_+^{-1}[\varphi''(T, \mu, T \cdot)] - \frac{\partial \nu_n(\mu, T)}{\partial \mu} \tilde{L}_+^{-1}[\tilde{\varphi}] = \frac{\partial \tilde{\varphi}}{\partial \mu} + c_1(\mu, T) \varphi', \quad (4.9)$$

for some constant $c_1(\mu, T)$. Note that the expressions $\tilde{L}_+^{-1}[\tilde{\varphi}], \tilde{L}_+^{-1}[\tilde{\varphi}']$ make sense, since $\tilde{\varphi}, \tilde{\varphi}'' \perp \tilde{\varphi}'$.

As we mentioned above, we can also take a derivative with respect to T in (4.8). We get

$$\tilde{L}_+ \left[\frac{\partial \tilde{\varphi}}{\partial T} \right] + 2\mu^2 T^{-3} \tilde{\varphi}'' + \frac{\partial \nu_n(\mu, T)}{\partial T} \tilde{\varphi} = 0.$$

Note; however,

$$\frac{\partial \tilde{\varphi}}{\partial T}(y) = \partial_T \varphi(T, \mu, Ty) + y \varphi'(T, \mu, Ty).$$

Consequently, (recall $\tilde{L}_+[\varphi'] = 0$),

$$\tilde{L}_+ \left[\frac{\partial \tilde{\varphi}}{\partial T} \right] = \tilde{L}_+[\partial_T \varphi(T, \mu; T \cdot)] - 2\mu^2 T^{-1} \varphi''(T, \mu, Ty).$$

Thus, we obtain

$$\tilde{L}_+[\partial_T \varphi] + \frac{\partial \nu_n(\mu, T)}{\partial T} \tilde{\varphi} = 0.$$

It follows that

$$\tilde{L}_+^{-1}[\tilde{\varphi}] = -\frac{\partial_T \varphi}{\frac{\partial \nu_n(\mu, T)}{\partial T}} + c_2(\mu, T) \tilde{\varphi}'. \quad (4.10)$$

Taking a dot product with $\tilde{\varphi}$ (in the Hilbert space $L^2[-1, 1]$!) yields

$$\begin{aligned} \langle \tilde{L}_+^{-1}[\tilde{\varphi}], \tilde{\varphi} \rangle &= -\frac{1}{\frac{\partial \nu_n(\mu, T)}{\partial T}} \int_{-1}^1 \partial_T \varphi(T, Ty) \varphi(T, Ty) dy \\ &= -\frac{1}{T \frac{\partial \nu_n(\mu, T)}{\partial T}} \int_{-T}^T \partial_T \varphi(T, z) \varphi(T, z) dz \end{aligned}$$

$$= -\frac{1}{2T \frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T [\|\varphi\|^2] - \varphi^2(T)].$$

Now, taking a dot product with $\varphi''(T, \mu, Ty)$ (again in $L^2[-1, 1]$) and performing integration by parts yields

$$\begin{aligned} \langle \tilde{L}_+^{-1}[\tilde{\varphi}], \varphi''(T, T \cdot) \rangle &= -\frac{1}{\frac{\partial \nu_n(\mu, T)}{\partial T}} \int_{-1}^1 \partial_T \varphi(T, Ty) \varphi''(T, Ty) dy \\ &= \frac{T}{\frac{\partial \nu_n(\mu, T)}{\partial T}} \int_{-1}^1 \partial_T \varphi'(T, Ty) \varphi'(T, Ty) dy \\ &= \frac{1}{\frac{\partial \nu_n(\mu, T)}{\partial T}} \int_{-T}^T \partial_T \varphi'(T, z) \varphi'(T, z) dz = \frac{1}{2 \frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T \|\varphi'\|^2 - (\varphi'(T))^2]. \end{aligned}$$

Finally, taking dot product in (4.9) with $\varphi''(T, \mu, Ty)$ in $L^2[-1, 1]$ results in

$$\begin{aligned} 2\mu \langle \tilde{L}_+^{-1}[\varphi''(T, T \cdot), \varphi''(T, T \cdot)] - \frac{\partial \nu_n(\mu, T)}{\partial \mu} \langle \tilde{L}_+^{-1}[\tilde{\varphi}], \varphi''(T, T \cdot) \rangle \\ = \langle \frac{\partial \tilde{\varphi}}{\partial \mu}, \varphi''(T, T \cdot) \rangle. \end{aligned}$$

For the right-hand side, we have

$$\begin{aligned} \langle \frac{\partial \tilde{\varphi}}{\partial \mu}, \varphi''(T, \mu, T \cdot) \rangle &= \int_{-1}^1 \partial_\mu \varphi(T, \mu, Ty) \varphi''(T, \mu, Ty) dy = \\ &= - \int_{-T}^T \partial_\mu \varphi'(T, \mu, z) \varphi'(T, \mu, z) dz = -\frac{1}{2} \partial_\mu [\|\varphi'\|^2]. \end{aligned}$$

This last identity, together with our formula for $\langle \tilde{L}_+^{-1}[\tilde{\varphi}], \varphi''(T, T \cdot) \rangle$ yield

$$\langle \tilde{L}_+^{-1}[\varphi''(T, T \cdot), \varphi''(T, T \cdot)] = \frac{1}{4\mu} \left[\frac{\frac{\partial \nu_n(\mu, T)}{\partial \mu}}{\frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T \|\varphi'\|^2 - (\varphi'(T))^2] - \partial_\mu [\|\varphi'\|^2] \right].$$

Going back to the corresponding quantities that we are interested in, namely $\langle L_+^{-1} \varphi, \varphi \rangle$, $\langle L_+^{-1} \varphi'', \varphi \rangle$, $\langle L_+^{-1} \varphi'', \varphi'' \rangle$, we conclude

$$\langle L_+^{-1} \varphi, \varphi \rangle = \frac{1}{T} \langle \tilde{L}_+^{-1} \tilde{\varphi}, \tilde{\varphi} \rangle = -\frac{1}{2T^2 \frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T \|\varphi\|^2 - \varphi^2(T)], \quad (4.11)$$

$$\langle L_+^{-1} \varphi'', \varphi \rangle = \frac{1}{T} \langle \tilde{L}_+^{-1} \varphi''(T, T \cdot), \varphi(T, T \cdot) \rangle = \frac{1}{2T \frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T \|\varphi'\|^2 - (\varphi'(T))^2], \quad (4.12)$$

$$\langle L_+^{-1}\varphi'', \varphi'' \rangle = \frac{1}{4T\mu} \left[\frac{\frac{\partial \nu_n(\mu, T)}{\partial \mu}}{\frac{\partial \nu_n(\mu, T)}{\partial T}} [\partial_T \|\varphi'\|^2 - (\varphi'(T))^2] - \partial_\mu [\|\varphi'\|^2] \right]. \quad (4.13)$$

Furthermore, for all our waves $\varphi'(T) = 0$ and including the values of the derivatives $\frac{\partial \nu_n(\mu, T)}{\partial \mu}$, $\frac{\partial \nu_n(\mu, T)}{\partial T}$ in the formulas (4.11), (4.12), (4.13), we obtain

$$\langle L_+^{-1}\varphi, \varphi \rangle = \frac{T(\mu^2 - 1)}{4\mu^2\pi^2n^2} [\partial_T \|\varphi\|^2 - \varphi^2(T)] \quad (4.14)$$

$$\langle L_+^{-1}\varphi'', \varphi \rangle = -\frac{T^2(\mu^2 - 1)}{4\mu^2\pi^2n^2} \partial_T \|\varphi'\|^2 \quad (4.15)$$

$$\langle L_+^{-1}\varphi'', \varphi'' \rangle = \frac{1}{4T\mu} \left[\frac{T}{\mu(\mu^2 - 1)} \partial_T \|\varphi'\|^2 - \partial_\mu \|\varphi'\|^2 \right]. \quad (4.16)$$

If we now rewrite the stability condition $\det(U) < 0$ by inserting the formulas (4.14), (4.15), (4.16), this reads

$$\begin{aligned} & \left(\frac{\|\varphi\|^2}{4} + \frac{T(\mu^2 - 1)(1 - \nu)}{4\mu^4\pi^2n^2} [\partial_T \|\varphi\|^2 - \varphi^2(T)] + (1 - \mu^2) \langle L_+^{-1}\varphi', \varphi' \rangle \right) \\ & \times \left(\frac{\|\varphi'\|^2}{4} + \frac{(1 - \mu^2)}{4T\mu} \left[\frac{T}{\mu(\mu^2 - 1)} \partial_T \|\varphi'\|^2 - \partial_\mu \|\varphi'\|^2 \right] + \frac{1 - \nu}{\mu^2} \langle L_+^{-1}\varphi', \varphi' \rangle \right) \\ & \leq \frac{(1 - \nu)(1 - \mu^2)}{\mu^2} \left(\langle L_+^{-1}\varphi', \varphi' \rangle + \frac{T^2(\mu^2 - 1)}{4\mu^2\pi^2n^2} \partial_T \|\varphi'\|^2 \right)^2. \end{aligned} \quad (4.17)$$

In order to express the stability condition (4.17) in terms of an inequality between two function of the relevant variables (say μ, κ), we need to find suitable formulas for the expression in (4.4), as well as $\varphi(T)$, $\|\varphi\|$, $\|\varphi'\|$ and $\partial_T \|\varphi\|^2$, $\partial_T \|\varphi'\|^2$, $\partial_\mu \|\varphi'\|^2$. Our remaining task is to find these formulas and state the appropriate results for the cases $p = 2, 3$.

4.3. The case $p = 2$. First we rewrite (2.7) in the form

$$\varphi(x) = \varphi_0(1 - \beta sn^2(\alpha x, \kappa)),$$

where $\beta = \frac{\varphi_0 - \varphi_1}{\varphi_0}$. From equalities (2.9), we have

$$\varphi_0 = 2\mu^2\alpha^2(1 + \kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4}), \quad \varphi_1 = 2\mu^2\alpha^2(1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4}).$$

It is convenient to express β in terms of κ only. Indeed, plugging in the formulas for φ_0, φ_1 , we have

$$\beta = \beta(\kappa) = \frac{\varphi_0 - \varphi_1}{\varphi_0} = \frac{3\kappa^2}{1 + \kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4}}. \quad (4.18)$$

By integrating (2.3) over one period, we obtain

$$\|\varphi\|^2 = \nu \int_{-T}^T \varphi dx,$$

and from (2.9), we get

$$\begin{aligned} \|\varphi\|^2 &= \frac{16\mu^4 K^3(\kappa)}{T^3} \left[3E(\kappa) \sqrt{1 - \kappa^2 + \kappa^4} + K(\kappa) \right. \\ &\quad \left. \times (\kappa^2 - 2 + \sqrt{1 - \kappa^2 + \kappa^4}) \sqrt{1 - \kappa^2 + \kappa^4} \right]. \end{aligned}$$

Denoting

$$\begin{aligned} G_{21}(\kappa) &:= K^3(\kappa) \left[3E(\kappa) \sqrt{1 - \kappa^2 + \kappa^4} + K(\kappa) \right. \\ &\quad \left. \times (\kappa^2 - 2 + \sqrt{1 - \kappa^2 + \kappa^4}) \sqrt{1 - \kappa^2 + \kappa^4} \right], \end{aligned}$$

we have

$$\|\varphi\|^2 = \frac{16\mu^4}{T^3} G_{21}(\kappa). \quad (4.19)$$

Next, since $\kappa^2 \leq \beta < 1$, then

$$\begin{aligned} &\int_0^{K(\kappa)} \frac{1}{[1 - \beta s n^2(y)]^2} dy \\ &= \frac{\beta E(\kappa) - (\kappa^2 - \beta)K(\kappa) + (2\beta\kappa^2 + 2\beta - \beta^2 - 3\kappa^2)\Pi(\pi/2, \beta, \kappa)}{2(\beta - 1)(\kappa^2 - \beta)}, \end{aligned}$$

where Π is the elliptic integral of third kind. Since $\beta = \beta(\kappa)$ from (4.18), we denote

$$I(\kappa) := \frac{\beta E(\kappa) - (\kappa^2 - \beta)K(\kappa) + (2\beta\kappa^2 + 2\beta - \beta^2 - 3\kappa^2)\Pi(\pi/2, \beta, \kappa)}{2(\beta - 1)(\kappa^2 - \beta)}.$$

From (2.7), we have

$$\begin{aligned} \varphi(T) &= \varphi_1 = 2\mu^2 \alpha^2 (1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4}) \\ &= \frac{2\mu^2}{T^2} K^2(\kappa) (1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4}). \end{aligned} \quad (4.20)$$

Replacing the above expressions in (4.5) and (4.6), for the terms in the right side of (4.3) we obtain

$$I_{11} = \frac{4\mu^4}{T^3} \left[\frac{(1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^4}{2(1 + \kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^2} K^3(\kappa) I(\kappa) - G_{21}(\kappa) \right] \quad (4.21)$$

$$\varphi^2(T) I_{12} = \frac{4\mu^4}{T^3} (1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^2 K^3(\kappa) Q(\kappa) \quad (4.22)$$

$$\frac{\varphi(T)}{\phi(T)} I_{12}^2 = \frac{4\mu^4 (1 + \kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^2 K^3(\kappa)}{T^3 I(\kappa)} Q^2(\kappa), \quad (4.23)$$

where

$$Q(\kappa) := \frac{(1 - 2\kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^2}{(1 + \kappa^2 + \sqrt{1 - \kappa^2 + \kappa^4})^2} I(\kappa) - K(\kappa).$$

Plugging these three formulas in (4.4) yields the expression for $\langle L_-^{-1}[\varphi'], \varphi' \rangle$.

Differentiating (4.19) with respect to T , we get

$$\partial_T ||\varphi||^2 = 16\mu^4 \frac{T^3 \frac{dG_{21}(\kappa)}{d\kappa} \frac{d\kappa}{dT} - 3T^2 G_{21}(\kappa)}{T^6}.$$

From (2.9) and then (2.8), we have

$$\nu = 4\mu^2 \alpha^2 \sqrt{1 - \kappa^2 + \kappa^4} = 4\mu^2 \frac{\sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa)}{T^2},$$

combining this with (4.7), yields the relation

$$1 + \frac{\mu^2}{\mu^2 - 1} \left(\frac{\pi n}{T} \right)^2 = 4\mu^2 \frac{\sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa)}{T^2}. \quad (4.24)$$

This gives an useful formula for T in terms of μ, κ, n , namely,

$$T = \mu \sqrt{4\sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa) + \frac{\pi^2 n^2}{1 - \mu^2}}. \quad (4.25)$$

Differentiating (4.24) with respect to T and solving for $\frac{d\kappa}{dT}$ yields

$$\frac{d\kappa}{dT} = \frac{\kappa(1 - \kappa^2)\sqrt{1 - \kappa^2 + \kappa^4} \left[2\sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa) - \frac{n^2 \pi^2}{2(\mu^2 - 1)} \right]}{T[2(1 - \kappa^2 + \kappa^4)E(\kappa)K(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K^2(\kappa)]}. \quad (4.26)$$

Replacing in the equation for $\partial_T ||\varphi||^2$, we get

$$\partial_T ||\varphi||^2 = \frac{16\mu^4}{T^4} \left[-G_{22} \frac{n^2 \pi^2}{2(\mu^2 - 1)} - G_{23} \frac{n^2 \pi^2}{2(\mu^2 - 1)} + G_{24} - G_{25} + G_{26} \right], \quad (4.27)$$

where

$$G_{22} = 3E(\kappa)K(\kappa) + (\kappa^2 - 2 + 2\sqrt{1 - \kappa^2 + \kappa^4})K^2(\kappa)$$

$$G_{23} = \frac{(1 - \kappa^2 + \kappa^4)K(\kappa)[3E^2(\kappa) - 2(2 - \kappa^2)E(\kappa)K(\kappa) + (1 - \kappa^2)K^2(\kappa)]}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)}$$

$$G_{24} = \frac{2(1 - \kappa^2 + \kappa^4)^{\frac{3}{2}} K^3(\kappa)[3E^2(\kappa) - 2(2 - \kappa^2)E(\kappa)K(\kappa) + (1 - \kappa^2)K^2(\kappa)]}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)}$$

$$G_{25} = \sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa)[3E(\kappa)K(\kappa) - (2 - \kappa^2)K^2(\kappa)]$$

$$G_{26} = (1 - \kappa^2 + \kappa^4)K^4(\kappa).$$

Concerning $\|\varphi'\|_{L^2}^2$, we have

$$\begin{aligned} \|\varphi'\|^2 &= \frac{96}{5T^5} \mu^4 K^5(\kappa) [2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)] \\ &= \frac{96\mu^4}{5T^5} G_{31}(\kappa), \end{aligned} \quad (4.28)$$

and

$$\partial_T \|\varphi'\|^2 = \frac{96}{5} \mu^4 \left[\frac{T^5 \frac{d}{d\kappa} G_{31}(\kappa) \frac{d\kappa}{dT} - 5T^4 G_1(\kappa) \right]. \quad (4.29)$$

From (4.26), we get

$$\partial_T \|\varphi'\|^2 = \frac{96\mu^4}{5T^6} \left[G_{32} + G_{33} \frac{n^2 \pi^2}{2(1 - \mu^2)} \right], \quad (4.30)$$

where

$$\begin{aligned} G_{32} &= \frac{15\kappa^4(1 - \kappa^2)^2 K^7(\kappa)}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)} \\ G_{33} &= \frac{10\sqrt{1 - \kappa^2 + \kappa^4} K^3(\kappa)}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)} \\ &\quad \times [(1 - \kappa^2 + \kappa^4)E^2(\kappa) + (1 - \kappa^2)^2 K^2(\kappa) - (1 - \kappa^2)(2 - \kappa^2)E(\kappa)K(\kappa)]. \end{aligned}$$

Differentiating (4.28) with respect to μ , we obtain

$$\partial_\mu \|\varphi'\|^2 = \frac{96}{5T^5} \left[4\mu^3 G_{31}(\kappa) + \mu^4 \frac{dG_{31}(\kappa)}{d\kappa} \frac{d\kappa}{d\mu} \right]. \quad (4.31)$$

Differentiating (4.24) with respect to μ and solving for $\frac{d\kappa}{d\mu}$ yields

$$\begin{aligned} \frac{d\kappa}{d\mu} &= \frac{\kappa(1 - \kappa^2)\sqrt{1 - \kappa^2 + \kappa^4}}{\mu[2(1 - \kappa^2 + \kappa^4)E(\kappa)K(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K^2(\kappa)]} \\ &\quad \times \left[2\sqrt{1 - \kappa^2 + \kappa^4} K^2(\kappa) - \frac{n^2 \pi^2}{2(\mu^2 - 1)^2} \right]. \end{aligned} \quad (4.32)$$

Replacing (4.32) in (4.31) yields

$$\partial_\mu \|\varphi'\|^2 = \frac{96\mu^3}{5T^5} \left[-G_{41} \frac{n^2 \pi^2}{2(\mu^2 - 1)} + G_{42} + G_{43} \right], \quad (4.33)$$

where

$$\begin{aligned} G_{41} &= \frac{10\sqrt{1 - \kappa^2 + \kappa^4} K^3(\kappa)}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)} \\ &\quad \times [(1 - \kappa^2 + \kappa^4)E^2(\kappa) + (1 - \kappa^2)^2 K^2(\kappa) - (1 - \kappa^2)(2 - \kappa^2)E(\kappa)K(\kappa)] \end{aligned}$$

$$\begin{aligned}
G_{42} &= \frac{20(1 - \kappa^2 + \kappa^4)K^5(\kappa)}{2(1 - \kappa^2 + \kappa^4)E(\kappa) - (1 - \kappa^2)(2 - \kappa^2)K(\kappa)} \\
&\quad \times [(1 - \kappa^2 + \kappa^4)E^2(\kappa) + (1 - \kappa^2)^2K^2(\kappa) - (1 - \kappa^2)(2 - \kappa^2)E(\kappa)K(\kappa)] \\
G_{43} &= 8(1 - \kappa^2 + \kappa^4)E(\kappa)K^5(\kappa) - 4(1 - \kappa^2)(2 - \kappa^2)K^6(\kappa).
\end{aligned}$$

We are now ready to formulate the main result for the case $p = 2$.

Theorem 2. *Let $\omega, c > 0 : \omega^2 + c^2 < 1$ and φ is the cnoidal solution displayed in (2.7). This solution is spectrally stable with respect to perturbations of the same period if and only if⁶ the condition (4.17) is satisfied. All quantities involved in (4.17) are functions of the independent continuous variables $\kappa, \mu \in (0, 1)$ and the discrete parameter n .*

We need to explain this last statement in Theorem 2. Indeed, formula (4.20) yields the expression for $\varphi(T)$, combining formulas (4.21), (4.22) and (4.23) yields $\langle L_-\varphi', \varphi' \rangle$, (4.27) presents $\partial_T \|\varphi\|^2$. In addition, (4.30) yields a formula for $\partial_T \|\varphi'\|^2$, (4.33) - for $\partial_\mu \|\varphi'\|^2$, (4.19) for $\|\varphi\|^2$ and finally, (4.28) for $\|\varphi'\|^2$. In all this formulas, there are the parameters κ, μ, n, T , but of course T could be expressed in terms of the first three via formula (4.25). An immediate corollary of the asymptotics of (4.17) is the following.

Corollary 1. *As $\mu \rightarrow 0+$, the cnoidal waves are spectrally unstable.*

Proof. One needs to go over the formulas for various expressions to conclude that the left hand side of (4.17) has the asymptotic $C(\kappa, n)\mu^{-7} + O(\mu^{-6})$, $C(\kappa, n) > 0$, while the right hand side is $O(\mu^{-6})$. Thus, for $0 < \mu \ll 1$, the inequality (4.17) fails and thus, the wave is spectrally unstable. $\square$

4.4. Case $p=3$. From (2.10) and (2.11), we have

$$\varphi(T) = \sqrt{2\nu} \sqrt{\frac{1-\kappa^2}{2-\kappa^2}}. \quad (4.34)$$

Using that

$$\int_0^{K(\kappa)} dn^2(x) dx = E(\kappa) \quad \text{and} \quad \int_0^{K(\kappa)} \frac{1}{dn^2(x)} dx = \frac{E(\kappa)}{1 - \kappa^2},$$

we get

$$\phi(T) = \varphi(T) \frac{\sqrt{\mu}(2 - \kappa^2)^{3/2} E(\kappa)}{2(1 - \kappa^2)\nu^{3/2}}$$

⁶with the exception of the case when one has equality in (4.17)

$$I_{11} = -\sqrt{\nu\mu} \frac{\kappa^2 E(\kappa)}{\sqrt{2-\kappa^2}}, \quad I_{12} = \sqrt{\frac{\mu}{\nu}} \sqrt{2-\kappa^2} [E(\kappa) - K(\kappa)].$$

Replacing the above identities in (4.4), we obtain

$$\langle L_-^{-1}[\varphi'], \varphi' \rangle = \frac{\sqrt{\mu\nu}}{\mu^2} \frac{E^2(\kappa) - (1-\kappa^2)K^2(\kappa)}{\sqrt{2-\kappa^2}E(\kappa)}. \quad (4.35)$$

We have

$$\|\varphi\|^2 = \varphi_0^2 \int_{-T}^T dn^2(\alpha x) dx = \frac{2\varphi_0^2}{\alpha} \int_0^{K(\kappa)} dn^2(y) dy = 4\mu^2 \frac{E(\kappa)K(\kappa)}{T}, \quad (4.36)$$

and

$$\partial_T \|\varphi\|^2 = 4\mu^2 \partial_T \left(\frac{E(\kappa)K(\kappa)}{T} \right) = 4\mu^2 \frac{\frac{d}{d\kappa} (E(\kappa)K(\kappa)) \frac{d\kappa}{dT} T - E(\kappa)K(\kappa)}{T^2}.$$

Again, we need to compute $\frac{d\kappa}{dT}$. From (2.11), we have $\nu = \mu^2 \frac{(2-\kappa^2)K^2(\kappa)}{T^2}$. We resort to a formula similar to (4.24). We have from the previous equality and (4.7),

$$1 + \frac{\mu^2}{\mu^2 - 1} \left(\frac{\pi n}{T} \right)^2 = \mu^2 \frac{(2-\kappa^2)K^2(\kappa)}{T^2}. \quad (4.37)$$

Differentiating (4.37) with respect to T yields

$$\frac{d\kappa}{dT} = \frac{\kappa(1-\kappa^2)[(\mu^2-1)(2-\kappa^2)K^2(\kappa) - \pi^2 n^2]}{T(\mu^2-1)[(2-\kappa^2)E(\kappa)K(\kappa) - 2(1-\kappa^2)K^2(\kappa)]}.$$

Plugging in the formula for $\frac{d\kappa}{dT}$ in the expression for $\partial_T \|\varphi\|^2$ gives us

$$\partial_T \|\varphi\|^2 = \frac{4\mu^2}{T^2} \left[(2-\kappa^2)K^2(\kappa) \frac{F_1}{F_2} - \frac{n^2 \pi^2}{\mu^2 - 1} \frac{F_1}{F_2} - E(\kappa)K(\kappa) \right], \quad (4.38)$$

where

$$\begin{aligned} F_1 &= F_1(\kappa) = E^2(\kappa) - (1-\kappa^2)K^2(\kappa) \\ F_2 &= F_2(\kappa) = (2-\kappa^2)E(\kappa)K(\kappa) - 2(1-\kappa^2)K^2(\kappa). \end{aligned}$$

Using that

$$\int_0^{K(\kappa)} dn^4(y) dy = \frac{4-2\kappa^2}{3} E(\kappa) + \frac{\kappa^2-1}{3} K(\kappa),$$

we get

$$\|\varphi'\|^2 = \frac{4\mu^2}{3T^3} K^3(\kappa) [(2-\kappa^2)E(\kappa) - 2(1-\kappa^2)K(\kappa)] = \frac{4\mu^2}{3T^3} K^2(\kappa) F_2, \quad (4.39)$$

and

$$\begin{aligned}\partial_T \|\varphi'\|^2 &= \frac{4\mu^2}{3T^6} \left[\frac{d}{d\kappa} [(2 - \kappa^2)E(\kappa)K^3(\kappa) - 2(1 - \kappa^2)K^4(\kappa)] \frac{d\kappa}{dT} T^3 \right. \\ &\quad \left. - 3T^2 [(2 - \kappa^2)E(\kappa)K^3(\kappa) - 2(1 - \kappa^2)K^4(\kappa)] \right] \\ &= \frac{4\mu^2}{3T^4} \left[(2 - \kappa^2)K^2(\kappa) \frac{F_4}{F_2} - \frac{n^2\pi^2}{\mu^2 - 1} \frac{F_4}{F_2} - 3K^2(\kappa)F_2 \right],\end{aligned}\quad (4.40)$$

where $F_4 = F_4(\kappa) = 3(2 - \kappa^2)E^2(\kappa)K^2(\kappa) - 12(1 - \kappa^2)E(\kappa)K^3(\kappa) + 3(1 - \kappa^2)(2 - \kappa^2)K^4(\kappa)$. We have

$$\begin{aligned}\partial_\mu \|\varphi'\|^2 &= \frac{4}{3T^3} \left[2\mu [(2 - \kappa^2)E(\kappa)K^3(\kappa) - 2(1 - \kappa^2)K^4(\kappa)] \right. \\ &\quad \left. + \mu^2 \frac{d}{d\kappa} [(2 - \kappa^2)E(\kappa)K^3(\kappa) - 2(1 - \kappa^2)K^4(\kappa)] \frac{d\kappa}{d\mu} \right].\end{aligned}$$

Differentiating (4.37) with respect to μ implies

$$\frac{d\kappa}{d\mu} = -\frac{2}{\mu(\mu^2 - 1)} \frac{n^2\pi^2 + (\mu^2 - 1)^2(2 - \kappa^2)K^2(\kappa)}{\frac{d}{d\kappa} [(2 - \kappa^2)K^2(\kappa)]},$$

and

$$\partial_\mu \|\varphi'\|^2 = \frac{4\mu}{3T^3} \left[2K^2(\kappa)F_2 - \frac{n^2\pi^2}{(\mu^2 - 1)^2} \frac{F_4}{F_2} - (2 - \kappa^2)K^2(\kappa) \frac{F_4}{F_2} \right]. \quad (4.41)$$

Note that from (4.37), we have the following formula

$$T = T(\kappa, \mu; n) = \mu \sqrt{\frac{\pi^2 n^2}{1 - \mu^2} + (2 - \kappa^2)K^2(\kappa)}. \quad (4.42)$$

Thus, we have proved the following result.

Theorem 3. *Let $\omega, c > 0 : \omega^2 + c^2 < 1$ and φ be the dnoidal solution (2.10) to the Klein-Gordon equation (1.1) for $p = 3$. Then, the solution is spectrally stable, with respect to perturbations of the same period if and only if the inequality (4.17) is satisfied.*

Remark. The concrete expression for the condition (4.17) in terms of κ, μ, n will take several pages, but note that all quantities in (4.17) are functions of the continuous parameters $\mu, \kappa \in (0, 1)$ and the discrete parameter $n \in \mathbb{Z}$ only. More precisely, (4.34) gives the expression for $\varphi(T)$, (4.35) for $\langle L - \varphi', \varphi' \rangle$, (4.36) for $\|\varphi\|^2$, (4.38) for $\partial_T \|\varphi\|^2$, (4.39) for $\|\varphi'\|^2$, (4.40) for $\partial_T \|\varphi'\|^2$, (4.41) for $\partial_\mu \|\varphi'\|^2$ - all in terms of κ, μ, n . Based on the formulas for the involved quantities, we have the following.

Corollary 2. *There exists $\epsilon > 0$, so that for all $\mu : 0 < \mu < \epsilon$, the corresponding waves are spectrally unstable.*

Proof. It suffices to look at the asymptotics of expression (4.17) as $\mu \rightarrow 0+$.

More specifically, as $\mu \rightarrow 0+$, we have that the left hand side of (4.17) behaves like $C(T, n)\mu^{-13/2} + O(\mu^{-6})$ for some positive $C(T, n)$, while the right hand side has the form $C_1(T, n)\mu^{-5} + O(\mu^{-9/2})$. Thus, (4.17) is violated for $\mu \ll 1$ and instability follows. $\square$

4.5. The case $p = 5$. In this case, due to the complexity of the formulas we will restrict ourselves to the case $c = 0$, which simplifies the computations quite a bit.

With this restriction $\mu = 1$, $\nu = 1 - \omega^2$ and the matrix U reduced to the diagonal matrix

$$U := \begin{pmatrix} 1 + \langle \mathcal{H}^{-1} \mathcal{J} \psi_0, \mathcal{J} \psi_0 \rangle & 0 \\ 0 & 1 + \langle \mathcal{H}^{-1} \mathcal{J} \psi_1, \mathcal{J} \psi_1 \rangle \end{pmatrix},$$

and

$$\begin{aligned} \langle \mathcal{H}^{-1} \mathcal{J} \psi_0, \mathcal{J} \psi_0 \rangle &= 4 \|\varphi\|^{-2} \omega^2 \langle L_+^{-1}[\varphi], \varphi \rangle, \\ \langle \mathcal{H}^{-1} \mathcal{J} \psi_1, \mathcal{J} \psi_1 \rangle &= 4 \|\varphi'\|^{-2} \omega^2 \langle L_-^{-1}[\varphi'], \varphi' \rangle. \end{aligned}$$

Since the operator L_- is nonnegative, then $\langle L_-^{-1}[\varphi'], \varphi' \rangle \geq 0$ and it remains to determine the sign of $\langle L_+^{-1}[\varphi], \varphi \rangle$. Differentiating (2.3) (in the case $p = 5$) with respect to ν , yields

$$L_+^{-1} \varphi = -\frac{d\varphi}{d\nu}, \quad \langle L_+^{-1}[\varphi], \varphi \rangle = -\left\langle \frac{d\varphi}{d\nu}, \varphi \right\rangle = -\frac{1}{2} \frac{d}{d\nu} \int_{-T}^T \varphi^2 dx.$$

In [10] is shown that $\frac{d}{d\nu} \int_{-T}^T \varphi^2 dx < 0$, and thus $\langle L_+^{-1}[\varphi], \varphi \rangle > 0$. Thus, similar to [10], we arrive at the same condition that determines the stability of the wave. Thus, we have reproved the result in [10], which shows instability of the waves, when they satisfy the spectral assumption⁷ (3.8).

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⁷As pointed out before (3.8) was confirmed numerically in [10] and also by our own numerical runs (see Figure 1) for all values of T, ω

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